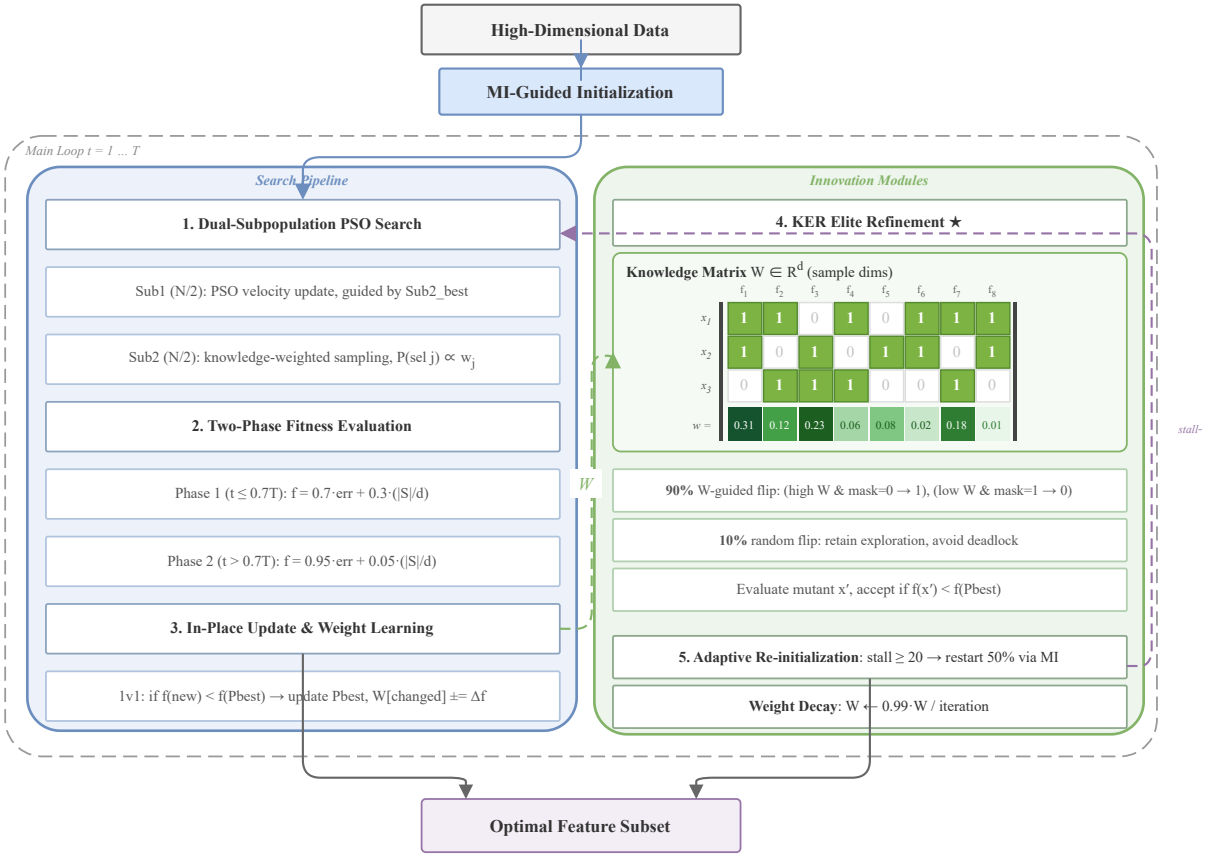

High-Dimensional Data
MI-Guided Initialization
Main Loop t = 1 ... T
Search Pipeline
1. Dual-Subpopulation PSO Search
Sub1 (N/2): PSO velocity update, guided by Sub2_best
Sub2 (N/2): knowledge-weighted sampling, P(sel j) ∝ w_j
2. Two-Phase Fitness Evaluation
Phase 1 (t ≤ 0.7T): f = 0.7·err + 0.3·(|S|/d)
Phase 2 (t > 0.7T): f = 0.95·err + 0.05·(|S|/d)
3. In-Place Update & Weight Learning
1v1: if f(new) < f(Pbest) → update Pbest, W[changed] ±= Δf
W
Innovation Modules
4. KER Elite Refinement ★
Knowledge Matrix W ∈ R^d (sample dims)
f1 f2 f3 f4 f5 f6 f7 f8
x1 1 1 0 1 0 1 1 1
x2 1 0 1 0 1 1 0 1
x3 0 1 1 1 0 0 1 0
w = 0.31 0.12 0.23 0.06 0.08 0.02 0.18 0.01
90% W-guided flip: (high W & mask=0 → 1), (low W & mask=1 → 0)
10% random flip: retain exploration, avoid deadlock
Evaluate mutant x′, accept if f(x′) < f(Pbest)
5. Adaptive Re-initialization: stall ≥ 20 → restart 50% via MI
Weight Decay: W ← 0.99·W / iteration
stall-
Optimal Feature Subset

→restart